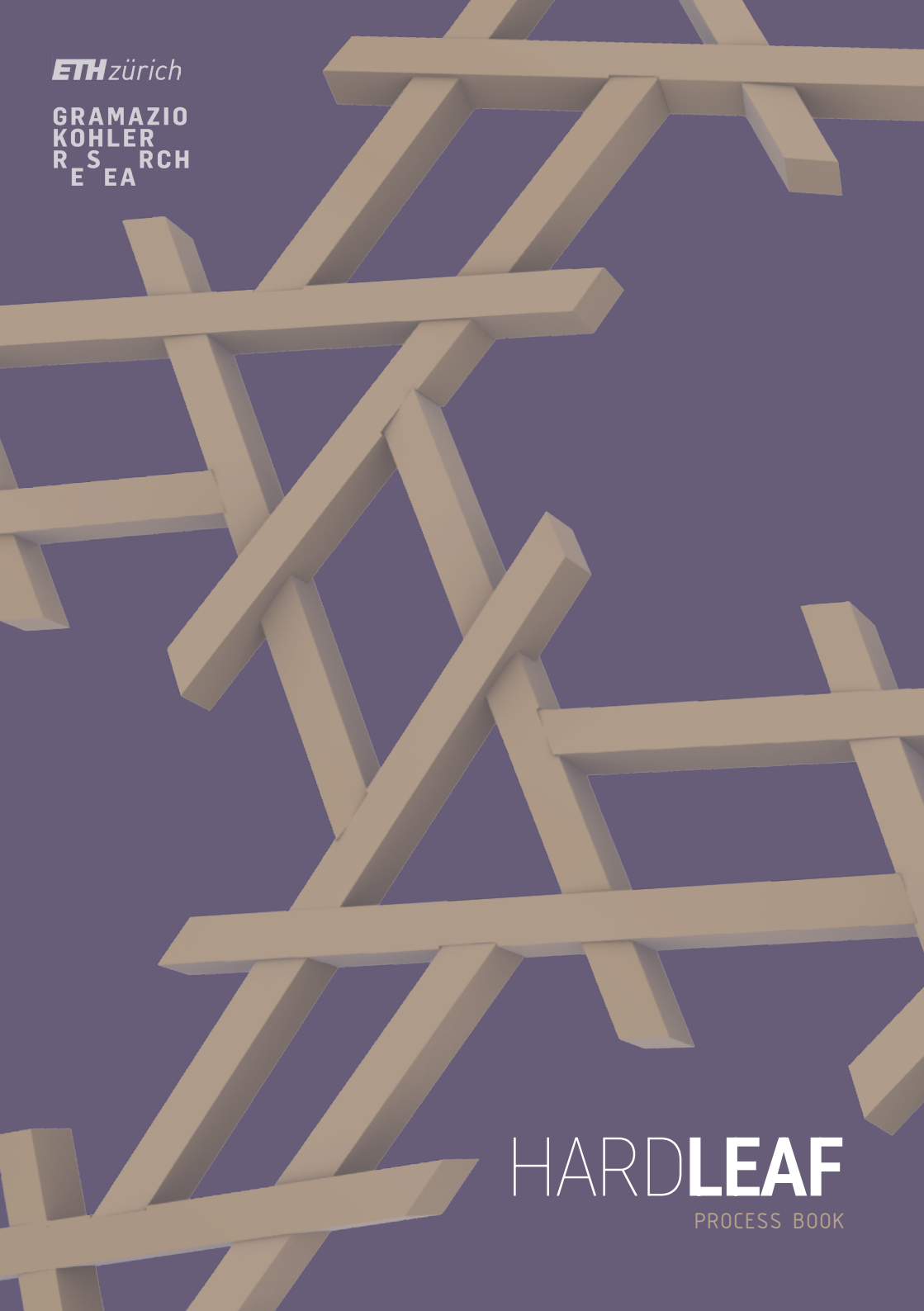




ETH zürich

GRAMAZIO
KOHLER
RESEARCH
E EA

HARDLEAF

PROCESS BOOK

GRAMAZIO
KOHLER
R S RCH
E EA

HARD**LEAF**

Quartiergarten Hard, Zurich

Web Site



C o n t e n t s

		page
01	Introduction	7
02	Context & Brief	9
03	Design - Proposal - Development - Final Design	17
04	Timeline	27
05	Process - Revision - Detailing - Fabrication - Assembly	35
06	Built Pavilion	52
07	Conclusion	60

I n t r o d u c t i o n

Computational design enables architectural systems to be developed through relationships between geometry, structure, material, and construction. Rather than defining each element individually, parameters and rules can be used to generate and adjust an interconnected system. This approach is particularly relevant for reciprocal-frame structures, in which each timber member is supported by neighbouring elements while simultaneously contributing to the stability of the overall network.

This project investigates the transition from such a computational system to a full-scale timber structure. The pavilion must not only satisfy a geometric idea, but also respond to structural behaviour, material dimensions, connection details, fabrication tolerances, transport, and assembly.

Rhino, Grasshopper, and Python are used to coordinate these dependencies and translate the architectural concept into a fabrication-ready timber model. Code is not understood solely as a tool for generating form, but as part of a continuous process connecting design, analysis, fabrication, and construction.

At the same time, the pavilion is designed as a framework for climbing vegetation. Kiwi plants will gradually grow across the structure and form a natural canopy, creating shade for the space below. The final architectural character will therefore emerge through the interaction between a precisely defined timber system, changing vegetation, weather, and everyday use.

2 / Context & Brief

C o n t e x t

The pavilion is located in Quartiergarten Hard, a community garden at Bullingerstrasse 90 in Zurich. The garden is used by a local association for cultivation, recreation, and neighbourhood activities. The new structure provides members with a shaded place for rest and gathering while serving as a permanent support for climbing plants.

The project was initiated during the spring semester as part of Coding Architecture II. Building on the computational foundations introduced in Coding Architecture I, the course focused on integrating architectural design with parametric modelling and programming. Students worked with Rhino and Grasshopper and developed Python-based tools through Visual Studio Code. During the first assignments, individual workflows were created for mesh generation, relaxation, geometric modification, and reciprocal-frame transformation. These components were later combined in a final group design project. Each group developed a proposal for Quartiergarten Hard (Fig. 1), starting from conceptual sketches, references, and spatial studies before translating it into a rule-based digital system.

The proposals addressed several interconnected questions. In addition to architectural and spatial quality, the groups considered structural stability, load-bearing behaviour, timber dimensions, joints, material use, prefabrication, and assembly. The reciprocal system could not be treated only as a formal pattern; it had to become a plausible and buildable timber structure. Throughout the semester, the projects were reviewed almost weekly with the teaching team, invited specialists, and members of the Quartiergarten association. The feedback led to repeated revisions and required the groups to balance computational experimentation, constructional feasibility, and the practical needs of future users. This balance was also central to the final selection process. While the course encouraged parametrically ambitious and spatially expressive proposals, the association prioritised shade, vegetation, safety, durability, and everyday usability. The teaching team and Quartiergarten representatives jointly selected one final proposal for further development. This winning design became the shared basis for the subsequent four-week Focus Work and its full-scale realisation.

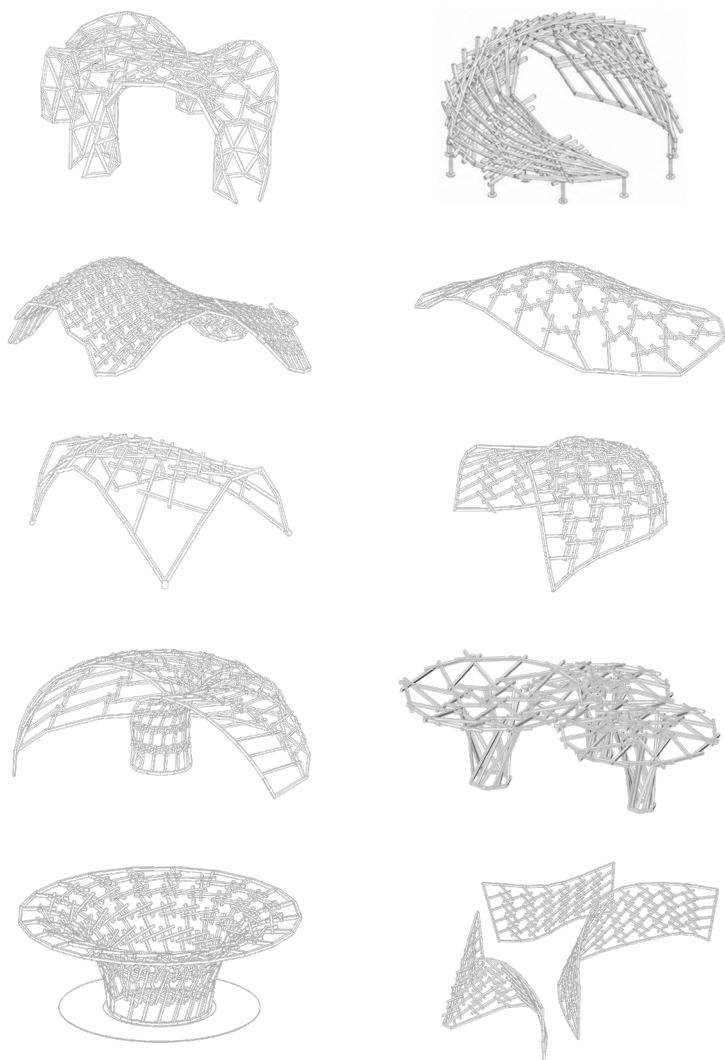


Fig. 1 - Different group proposals

B r i e f

The original task was to design a reciprocal timber pavilion for the Quartiergarten Hard in Zurich (Fig. 2, 3). The project combined architectural design, computational methods, and constructive logic and was intended from the beginning as a real building project rather than a purely speculative exercise. The proposal therefore had to be structurally plausible, materially coherent, and realistically fabricable and assemblable.

The pavilion was required to form an open and visually permeable spatial structure based on reciprocal-frame logic. It was to provide a place of shade, rest, and gathering that could be used throughout the year. Instead of a solid roof, shading was to be created primarily through kiwi plants growing across the timber framework. The geometry therefore had to support vegetation while maintaining a clear and accessible space below.

The design was limited to a maximum projected footprint of 40 m² and a maximum height of 3.0 metres. It also had to respond to the project perimeter, required distances from neighbouring parcels, and the existing vegetation. Trees and plants within the site were to be preserved wherever possible, making the relationship between the pavilion, the garden, and biodiversity part of the design strategy.

Material and construction constraints were integrated from an early stage. Approximately 300 linear metres of timber were available, including fourteen existing beams measuring 9 × 9 cm and 2.3 metres in length. The project was to use dry construction, mechanical joints with minimal metal components, and nature-friendly screw-pile foundations. Member dimensions, cutting waste, durability, prefabrication, transport, and on-site assembly therefore had to be considered as part of the computational model.



Fig. 2 – Site, Quartiergarten Hard, Zurich

Fig. 3 - General floorplan, Quartiergarten Hard, Zurich





Baumpatinnen:

- 1 Apfel: *Malus domestica* 'Zürichapfel'
Regina S.
- 2 Kaki: *Diospyros kaki* 'Tipo'
Susanne L.
- 3 Quitte: *Cydonia oblonga* 'Leskovac'
Christine A.
- 4 Apfel: *Malus domestica* 'Cox Orange'
Rhababergruppe, Nicole G.
- 5 Süßkirsche: *Prunus avium* 'Sunburst'
Shirana S.
- 6 Lotuspflaume: *Diospyros lotus*
Irene D.
- 7 Eberesche: *Sorbus aucuparia* (Strauchform)
Hühnergruppe, Claudia K.
- 8 Sauerkirsche: *Prunus cerasus* 'Schattenmorelle'
Karin S.
- 9 Apfelquitte: *Cydonia oblonga* 'Cydolus'
Susanne L.
- 10 Weinbergpfirsich: *Prunus persica* 'Weinbergpfirsich'
Nicole G.
- 11 Aprikose: *Prunus armeniaca*
In Abklärung
- 12 Pfirsich: *Prunus persica* 'Honey Sun'
Zuchttfeld, Theo L.
- 13 Minikiwi: *Actinidia arguta*
Vakant
- 14 Aprikose: *Prunus armeniaca*
Margret P.
- 15 Williamsbirne: *Pyrus communis* 'Williams Christ'
Zwetsche: *Prunus domestica*
Manuela S.
- 16 Olivenbäume: *Olea europaea*
Kräutergruppe
- 17 Granatapfel: *Punica granatum*
Milica K.

Bauminselgruppe:

- 1 Apfel: *Malus domestica* 'Berlepsch rot'
Claudia K.
- 2 Reineclaude: *Prunus domestica* 'Gr. gr. Reineclaude'
David A.
- 3 Zwetsche: *Prunus domestica* 'Davrovice'
Karin S.
- 4 Kletterobst-Pergola (versch. Kiwis, Reben, Beeren)
Milica K.
- 5 Japanische Faserbanane: *Musa basjoo*
Irene D.
- 6 Mirabelle: *Prunus domestica* 'Mirabelle de Nancy'
Kinderkollektiv, Bernadette T.
- 7 Aprikose: *Prunus armeniaca* 'Elsa'
Feige: *Ficus carica*
Martin T.
- 8 Schwarze Maulbeere: *Morus nigra* 'Wellington'
Bernadette T.

Wildlinge und Bäume Naturgarten

Bäume GSZ

Bäume, die von der Baumschnittgruppe beschnitten werden



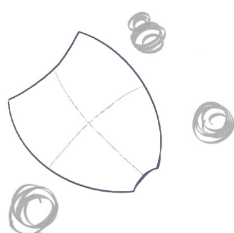




Fig. 4 – HardLeaf, initial proposal

The pavilion's core design concept (Fig. 4) was developed by Timo Feddern, Žan Kočunik, and Alessia Martino, three master's students in the Department of Architecture at ETH Zurich, as part of the course *Coding Architecture II*.

The design process focused on preserving and enhancing the existing vegetation of the garden. Particular attention was given to the plum tree and the banana trees, which emerged during the site visit as important and valued elements of the space. Rather than imposing a new object onto the site, the project was developed as an organic intervention integrated around the presence of these plants. As the design evolved, the geometry was refined to maximize the shaded area and improve environmental comfort. This led to the development of a leaf-shaped configuration, characterized by variations in height and a wide canopy capable of providing extensive coverage while maintaining a projected area of 40 m².

The pavilion retains a clear and simple structural logic based on two load-bearing sides. The covered area was intentionally designed as a wide and open space, ensuring flexibility and adaptability for different uses over time. The pavilion can accommodate tables, gatherings, workshops, or temporary installations, while remaining open to future transformations and changing configurations of plants within the garden.

The project also responds to the initial condition of the kiwi plants, which were located in the center of the site. Since relocating them immediately may not be possible, the project allows for multiple scenarios: the plants can temporarily remain within the open interior space or later be repositioned along the structural base, where they could continue growing integrated with the pavilion itself while keeping the central area unobstructed.

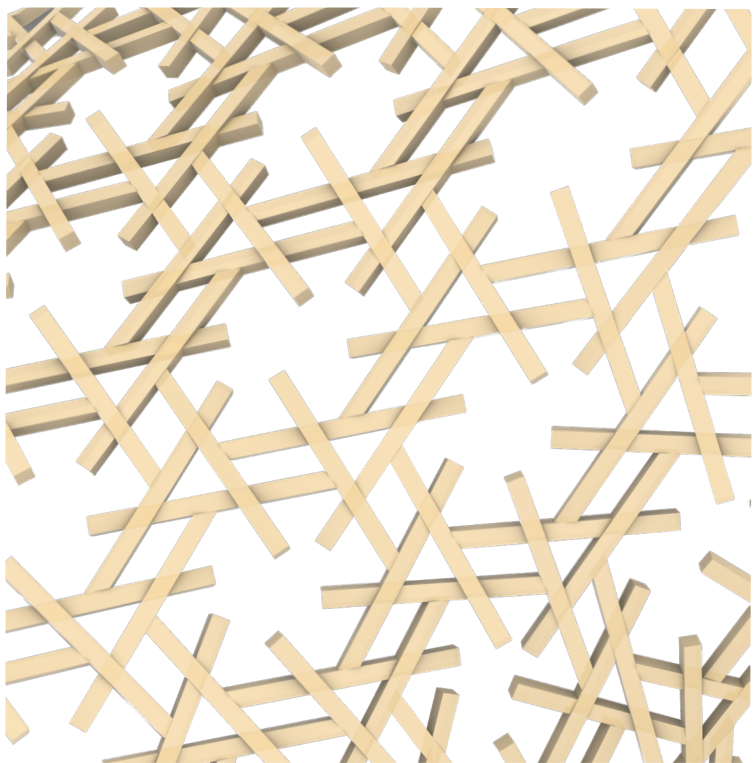


Fig. 5 - Reciprocal frame system

The reciprocal frame structure (Fig. 5) is generated through an organic meshing system derived from a hexagonal pattern (Fig. 6). Starting from a regular geometric grid, the mesh was progressively transformed and adapted to create a more fluid and natural configuration.

The resulting network combines order and irregularity: the hexagonal geometry provides structural continuity and stability, while the variation in orientation and density creates a lighter, more dynamic spatial effect.

The differentiated density of the mesh also allows the structure to accommodate multiple configurations of hanging plants and climbing vegetation, creating areas with varying levels of shade, openness, and interaction with the surrounding garden.

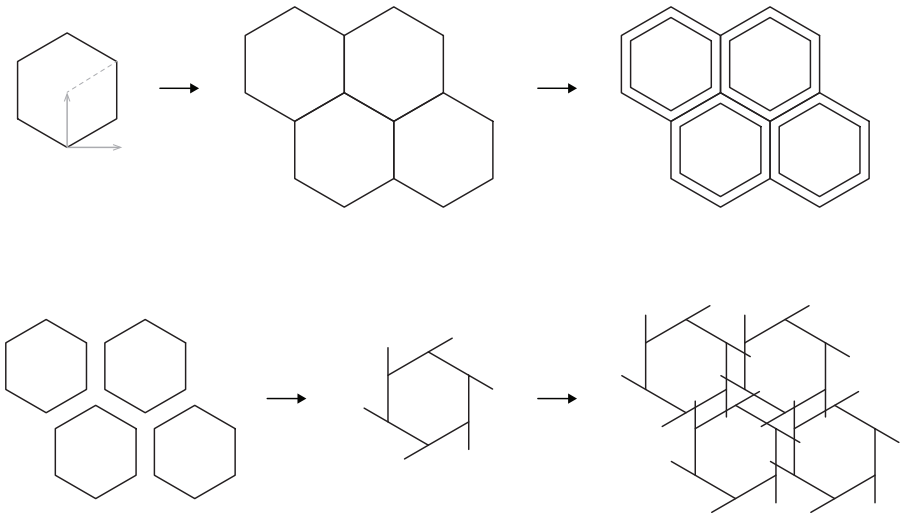


Fig. 6 - Workflow leading to final pattern

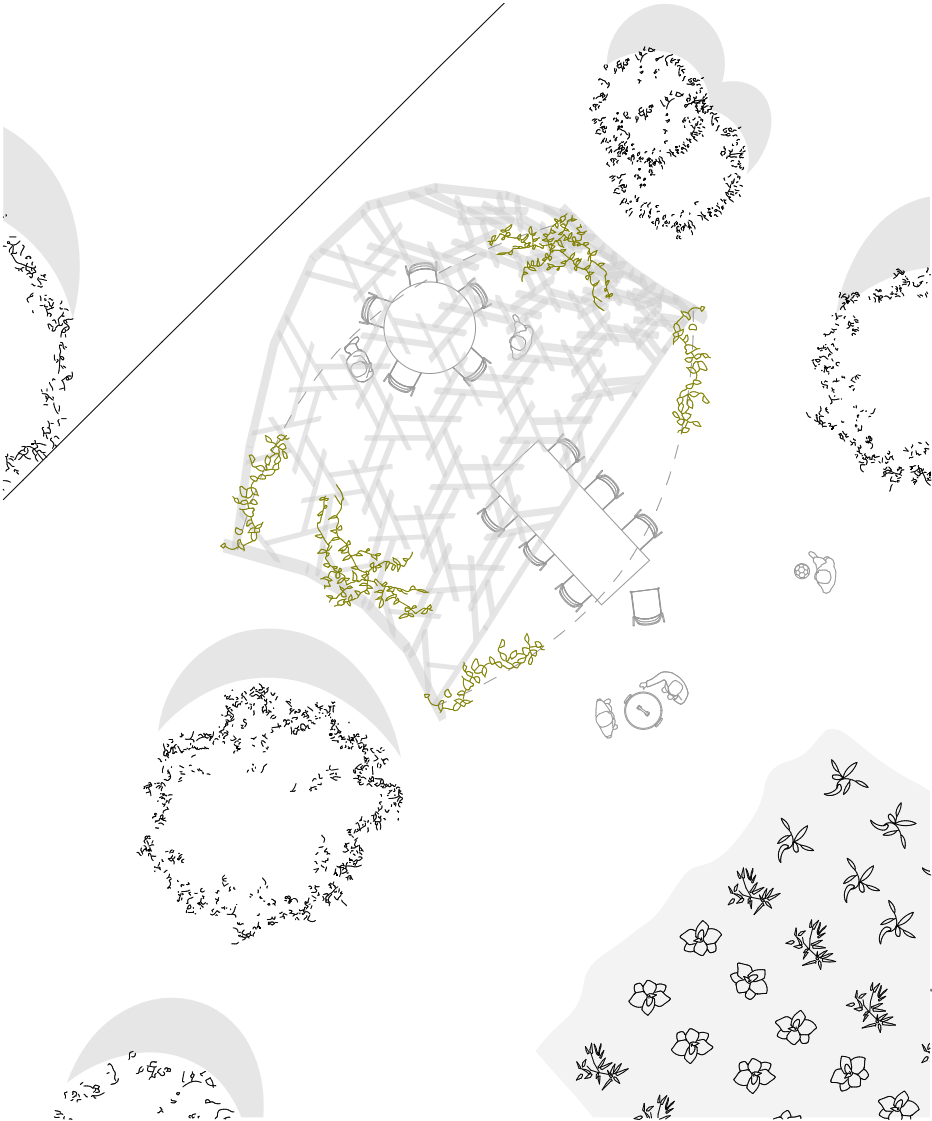


Fig. 7 - HardLeaf floorplan

To transform the initial concept into a design ready for fabrication, assembly, and safe installation on site, the project required a careful phase of technical refinement. This process involved addressing a series of practical and structural questions: How would the pavilion connect to the ground? How would the individual beams be joined? How could the components be efficiently categorized and fabricated? And how could the assembly process be made faster and more intuitive?

The pavilion was developed and constructed over four weeks in June 2026. The first two weeks were dedicated to design refinement, structural coordination, detailing, and fabrication preparation, while the final two weeks focused on component production, assembly, and on-site installation.

The work was carried out by 18 master's students from the Department of Architecture at ETH Zurich, under the coordination and supervision of the Coding Architecture II teaching team, with additional support from fabrication specialists and structural consultants.

At the same time, the design was further evaluated in relation to the needs of the Quartiergarten Hard community. The final orientation of the pavilion responds directly to their requests, particularly the need to maximise shading. The longer load-bearing edge is oriented towards the south, where it provides increased protection from direct solar exposure, while the shorter load-bearing side faces north. Between these two structural boundaries, the pavilion opens towards the garden, establishing a direct visual and spatial connection with the surrounding landscape.

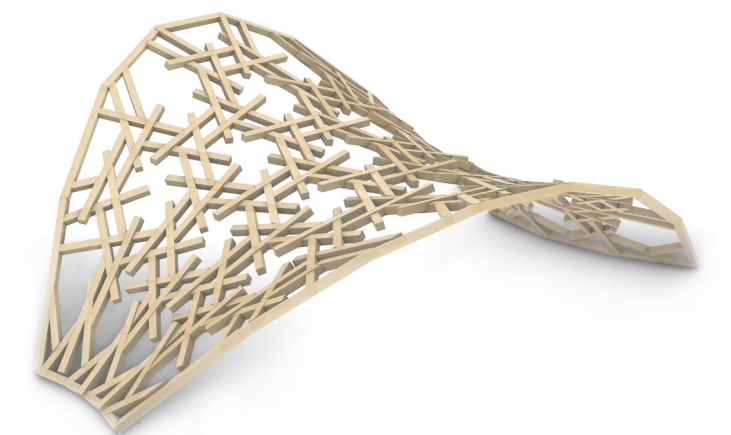


Fig. 8 - Initial design

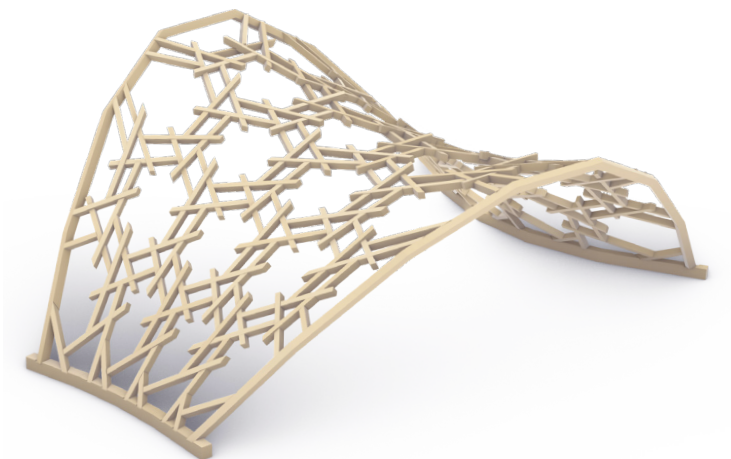


Fig. 9 - Final design

The original proposal (Fig. 8) defined the project's main spatial and architectural qualities, which were refined and technically resolved in the final design (Fig. 9) as a coordinated and buildable timber system. The overall geometry evolved to comply with dimension constraints, improve detailing, and provide sufficient space for the foundation connections. At the same time the structure was revised to create a more regular and efficient system, simplifying prefabrication, transport, and on-site assembly. The geometry was also optimized to reduce timber consumption and overall weight while preserving the intended spatial and structural qualities.

The project was organised around four main challenges: refinement of the computational model and design, technical detailing, fabrication, and assembly. Each phase was led by a group of four to five students, with responsibility for developing the relevant tasks and coordinating the required outputs. However, the phases were not treated as separate work packages. A collaborative structure with continuous exchange between groups was essential to maintaining consistency between the digital model and the built pavilion.

Although the reciprocal-frame geometry is inherently self-supporting, the timber members are additionally secured with mechanical fasteners to improve robustness, load transfer, and resistance to lateral actions such as wind. The pavilion is anchored through steel screw-pile foundations, allowing a dry, reversible installation with minimal disturbance to the garden soil.

The pavilion should not be understood as a finished and static object. As the climbing vegetation develops across the timber and cable system, its density, shade, and spatial atmosphere will continue to change. HardLeaf therefore combines a precisely defined computational structure with a living system whose final character will emerge gradually through growth, weathering, maintenance, and everyday use by the Quartiergarten community.

pic of people working oooor cute
group pic with everyone :)

Fig. 10 – Team work

Four-Week Workflow and Group Responsibilities

The Focus Work is organised as an intensive four-week process, covering 20 full working days from the final revision of the design to the construction of the pavilion and the concluding Apéro. The workflow is divided into four main phases: revision, preparation, fabrication and on-site assembly. Each phase builds directly on the results of the previous week, meaning that design decisions, digital models, fabrication data and construction planning must remain closely coordinated throughout the project.

The work is distributed across four groups (Fig. 10). Each group takes primary responsibility for one central phase: Group A leads Week 1, Group B Week 2, Group C Week 3 and Group D Week 4. Leading a week does not mean working independently. The responsible group coordinates the main tasks, defines priorities and ensures that the required results are completed on time. Once its main phase is finished, the group shifts toward supporting the following teams, solving remaining issues and contributing wherever additional capacity or specialist knowledge is needed. This creates a continuous handover rather than four isolated work packages.

Week 1: Revision and Detailing

Goal: Design freeze, coordinated timber model and fabrication-ready data

The first week focuses on finalising the design and establishing a reliable basis for fabrication. Feedback from the Quartiergarten community is integrated, while structural behaviour, stability, joints and construction details are reviewed. The digital timber model is coordinated so that geometry, dimensions and connections correspond to the intended structure. In parallel, the fabrication and assembly sequence, site logistics and foundation template are developed. By the end of the week, the main geometry should be frozen and ready for data extraction and production planning.

Week 2: Preparation and Machine Instructions

Goal: A complete system ready to enter fabrication

The second week translates the coordinated model into machine-readable production information. CNC instructions are prepared and tested through the BTLX and HOPS workflow, while the raw beams are pre-cut to their required stock lengths. The modular subdivision of the pavilion is also finalised, defining which elements can be prefabricated and how they will be transported and assembled. Remaining materials, fasteners and special components are checked and ordered so that fabrication can begin without major unresolved dependencies.

Week 3: Fabrication and Prefabrication

Goal: Completed and transport-ready timber modules

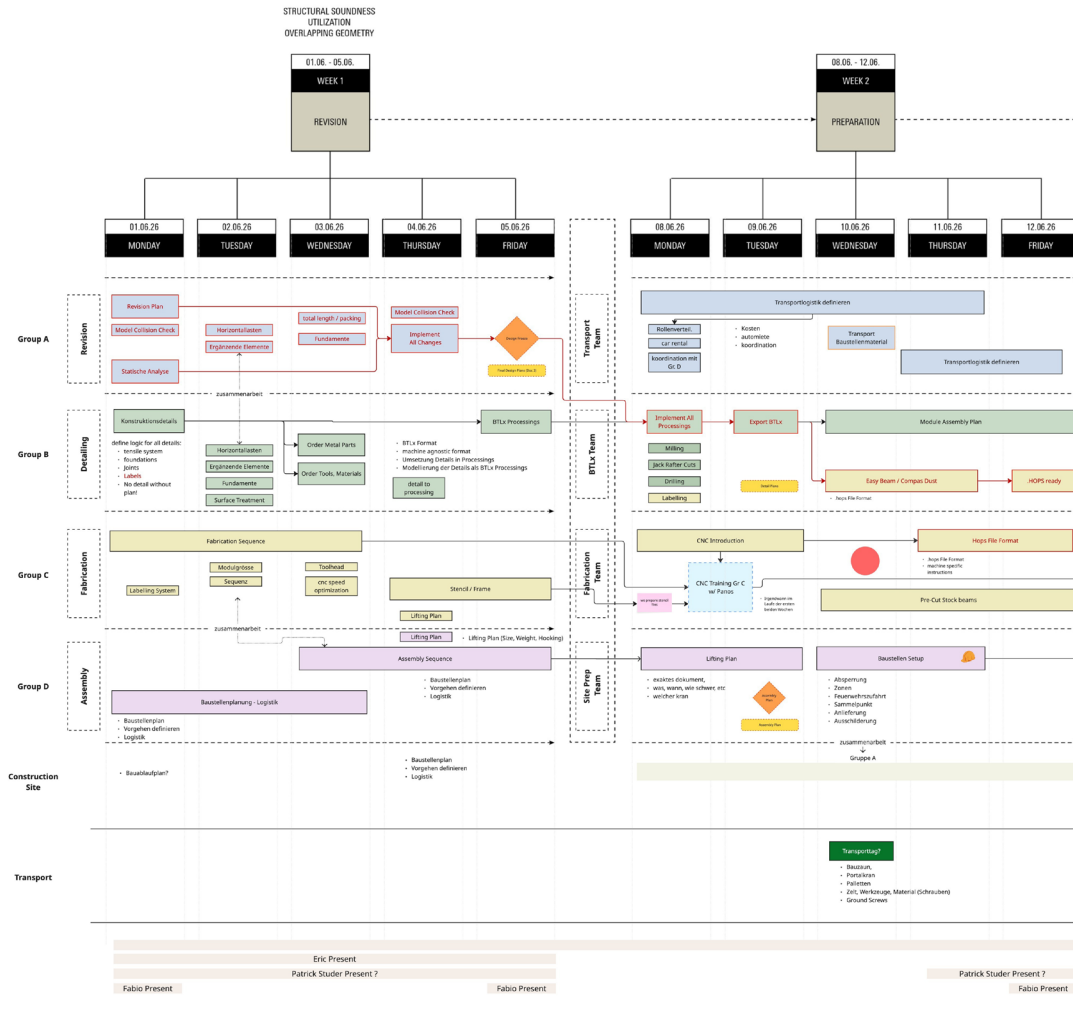
During the third week, the project moves from digital preparation to physical production. The foundation template and custom timber elements are manufactured according to the prepared machine instructions. Each component is checked, labelled and assigned to its correct position in the structure. The elements are then assembled into transportable modules in the workshop, allowing joints, tolerances and the planned assembly sequence to be tested in advance. All modules, loose parts, connectors and tools are subsequently organised and packed for transport.

Week 4: On-Site Assembly

Goal: Complete construction of the pavilion

The final week focuses on transport, foundation placement and full-scale construction at the Quartiergarten Hard. The foundation template is used to position the structure accurately, after which the prefabricated modules are assembled with crane support. Group D coordinates the site sequence, tools, lifting operations and communication between the teams, while the other groups contribute their knowledge from the earlier phases. After completion, the construction area is cleared, tools and remaining materials are organised, and the project concludes with the final handover and Apéro.

General Focus Work Plan



Preparation

• Revision Plan

• Standard System

• Standard System

• Standard System

• Standard System

Preparation

• Fabrication Sequence

• Assembly Sequence

• Assembly Sequence

• Assembly Sequence

• Assembly Sequence

Preparation

• BTLa Processings

• BTLa Processings

• BTLa Processings

• BTLa Processings

• BTLa Processings

Preparation

• Transportlogistik

• Transportlogistik

• Transportlogistik

• Transportlogistik

• Transportlogistik

Preparation

• BTLa Processings

• BTLa Processings

• BTLa Processings

• BTLa Processings

• BTLa Processings

Preparation

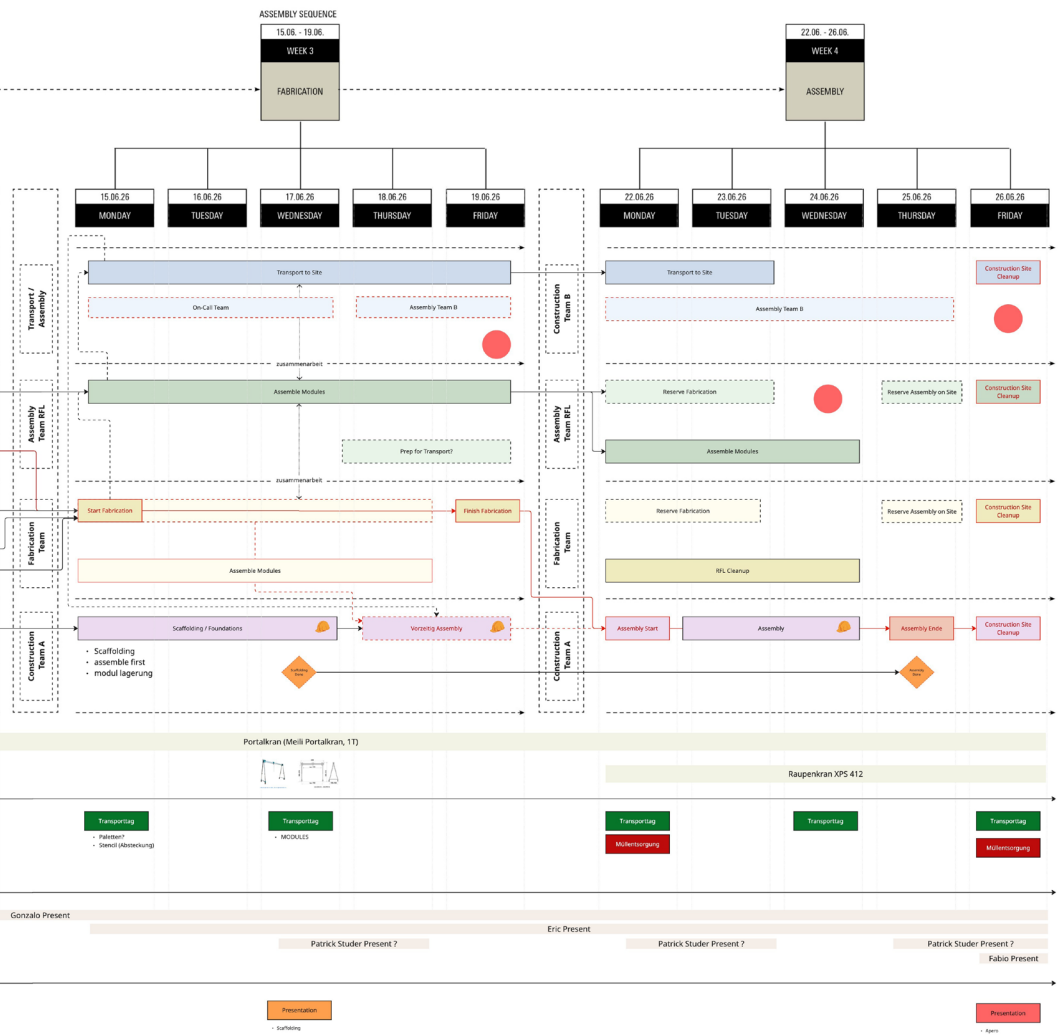
• Stand der Dinge?

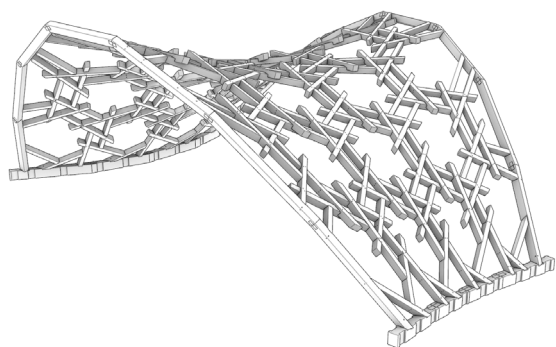
• Stand der Dinge?

• Stand der Dinge?

• Stand der Dinge?

• Stand der Dinge?





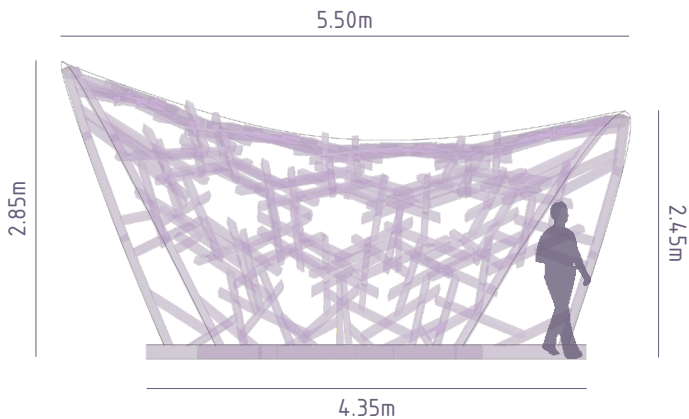


Fig. 11 – Elevation

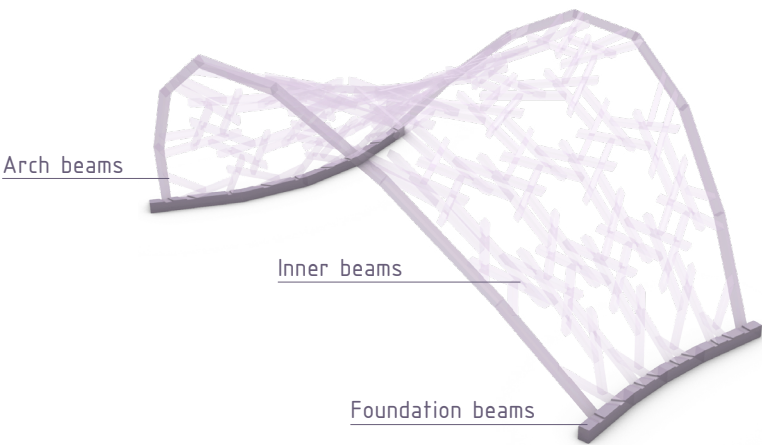


Fig. 12 – Beam Categorization

1. Responding to the Community Gardens Requirements

Following feedback from the Quartiergarten community, the pavilion geometry was adjusted to comply precisely with the maximum height requirement of 3 metres (Fig. 11). This revision also accounted for the ground-connection system, which raises the timber structure approximately 15 centimetres above ground level. The pavilion's lowest areas were also slightly raised to improve accessibility, usability, and spatial comfort beneath the structure.

2. Beam Categorisation

To allow the different structural elements to be controlled and detailed independently, the beams were divided into three main categories: outer arch beams, inner reciprocal-frame beams, and foundation beams (Fig. 12). This logical classification made it possible to assign specific cross-sections, orientations, joints, and fabrication rules to each part of the structure while maintaining a coordinated parametric model.

3. Orientation of Arch and Foundation Beams

The beam orientations were refined according to their structural and constructional roles. The centre lines of the foundation beams were aligned parallel to the ground plane, simplifying their connection to the screw-pile foundations. The cross-sections of the outer arch beams were oriented consistently within planes perpendicular to the ground, creating clearer and more reliable connection conditions between the arches and the inner reciprocal-frame beams.

4. Joint and Foundation Implementation

One of the main challenges was identifying suitable joints for each beam category. This process was developed in close coordination with the detailing team, Group B. Several alternatives were tested and compared according to their structural performance, visual appearance, material requirements, and ease of fabrication and assembly. Based on this evaluation, the most appropriate connection types were selected and integrated into the timber model code.

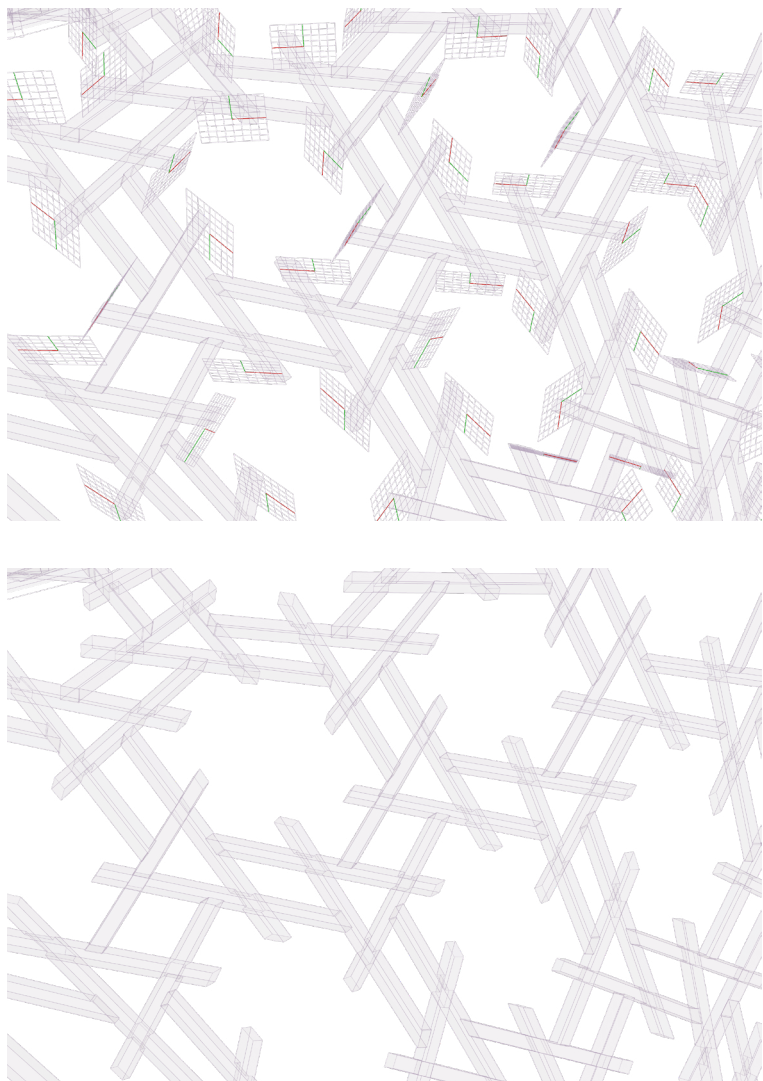


Fig. 13, 14 - Cutting planes, Final result with cleaned edges

5. Edge Clean-Up Using Cutting Planes

To create a cleaner and more coherent appearance along the pavilion's pattern, the exposed ends of the inner beams were trimmed using dedicated cutting planes (Fig. 13, 14). Each plane was aligned parallel to the corresponding edge of the underlying hexagonal mesh.

CODE LOGIC :

1. Identify Beams to Trim

Select inner beams with a TButt-type joint at one end and no end joint at the opposite, open end. XLap joints along the beam are ignored. Preview them through ``trimmed_inner_beams``.

2. Generate Cutting Planes

For each selected beam, find the nearest XLap joint to its open end. Use the other inner beam in this XLap as the reference beam. Create the cutting plane from the reference beam's centerline direction and local z-axis. Preview the planes through ``cutting_planes_inner_beams``.

3. Offset the Cutting Planes

Orient each plane normal toward the open end and move the plane along this normal using ``cutting_plane_inset_distance``. Positive values move toward the open end; negative values move in the opposite direction. The updated planes replace the original preview in ``cutting_planes_inner_beams``.

4. Trim the Beam Geometry

Apply a ``JackRafterCut`` using the finalized cutting plane. Remove the portion on the open-end side and retain the opposite portion. The cut is applied to the full beam solid, preserving the exact inclined intersection surface. The resulting geometry replaces the original beam in both ``trimmed_inner_beams`` and the final timber model.



Fig. 15 - Visualization of climbability solution

6. Structural Verification

The revised design was subsequently evaluated through structural analysis. The results were used to verify the overall stability and load-bearing behaviour of the pavilion and to identify areas requiring further adjustment. In particular, the geometry at the base was slightly modified to provide sufficient space for the screw-pile connections and to ensure that the foundation details could be implemented without interfering with the surrounding timber elements.

7. Climability and Vegetation Support

The pavilion is not intended to function as a climbing structure. Nevertheless, its open timber geometry and location within a public community garden, make it likely that accessible parts of the structure would be treated as climbable in practice, especially by children. For this reason, a non-climbable solution was preferred. Stainless-steel cables were introduced across the lower openings and mechanically fixed to the timber beams (Fig. 15, 16). These cables reduce the size of the accessible gaps and discourage climbing without visually closing the pavilion. At the same time, they provide an additional support system for climbing and hanging vegetation, allowing the safety intervention to contribute directly to the pavilion's long-term integration with the garden.

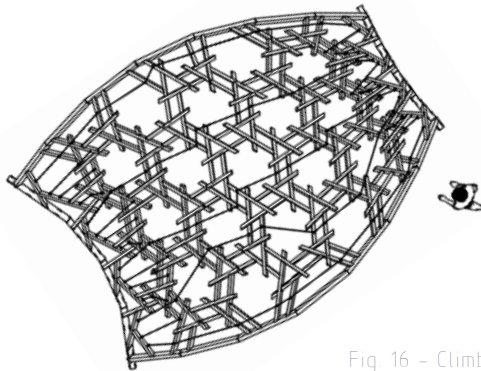
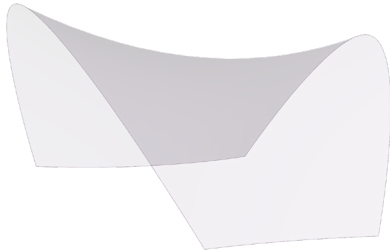


Fig. 16 - Climability solution plan

Grasshopper Workflow



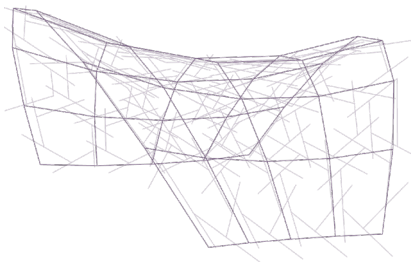
1 // PARAMETRIC SURFACE

A leaf-shaped surface is generated parametrically in Grasshopper and adjusted to meet the spatial and dimensional requirements of the community garden (maximum height: 3m; projected area: 40 m²).



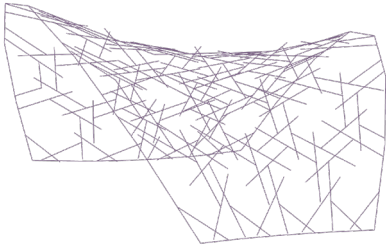
2 // HEXAGONAL MESH

The surface is discretized using a hexagonal pattern. Each hexagonal cell is then inset, creating a secondary network of smaller polygons. Their edges define the reference centerlines used to generate the timber beams.



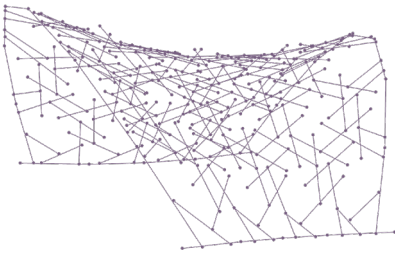
3 // RF SYSTEMS

Two reciprocal-frame systems are generated in parallel. The hexagonal mesh defines the inner beam network, while a simplified quad mesh generates the outer arches and foundation beams forming the structural perimeter.



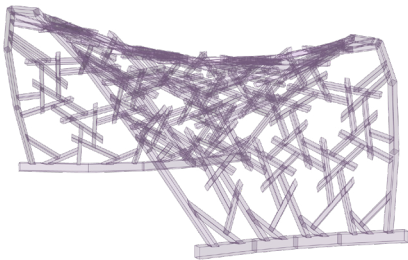
4 // HYBRID RF SYSTEM

A custom RF system merges the two networks into a single structural model. Inner beam centerlines are trimmed at their intersections with the outer arches to produce a clean integrated geometry.



5 // LIVE EDIT

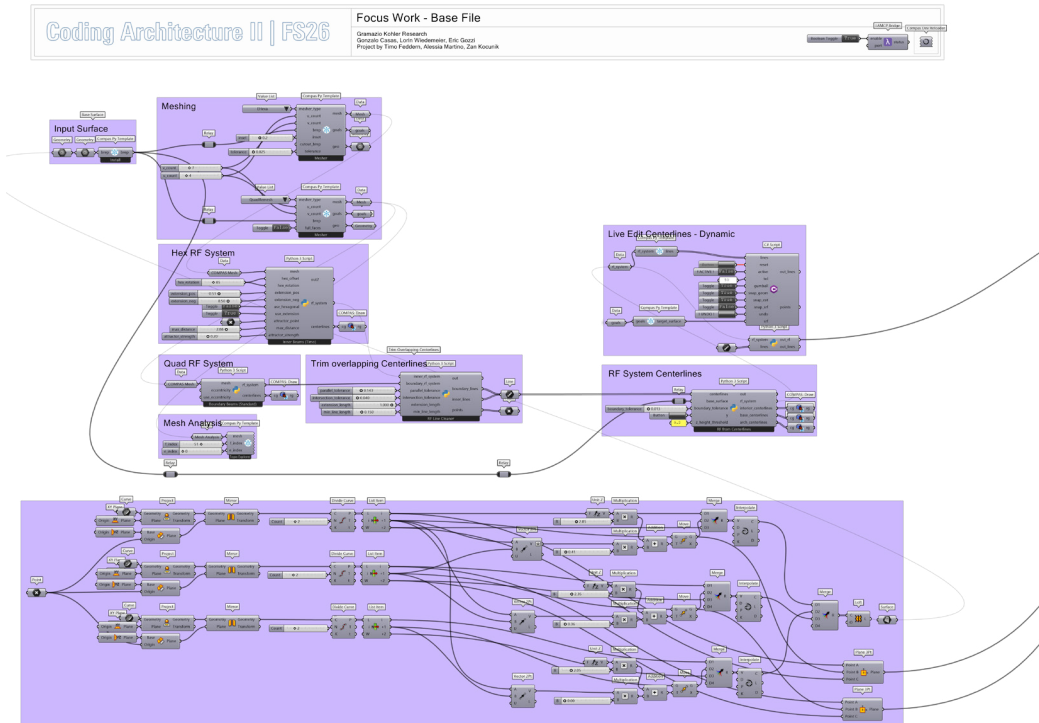
Live editing is used to resolve exceptional geometric conditions, such as missing joints, failed trims, or centerlines falling outside the defined tolerances. It also allows for targeted aesthetic adjustments.

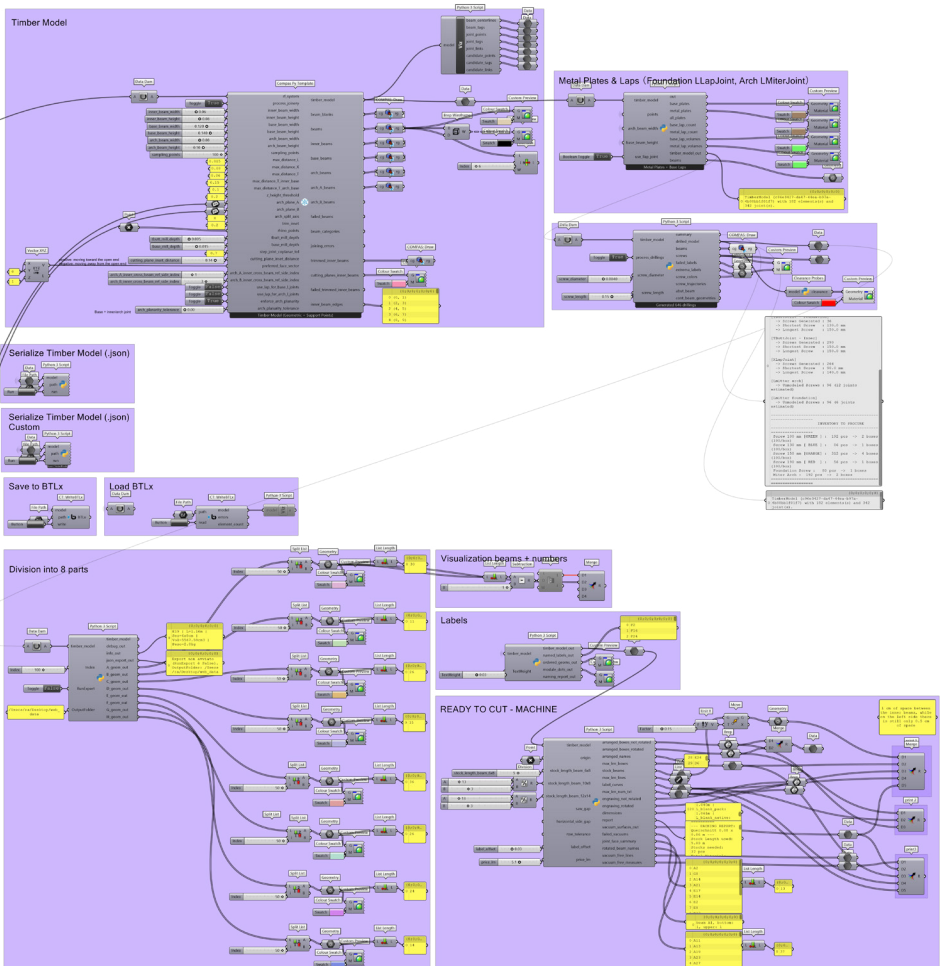


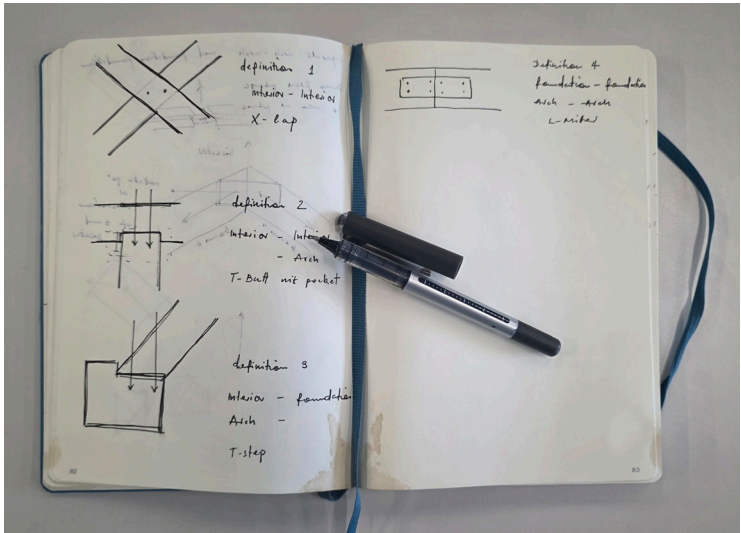
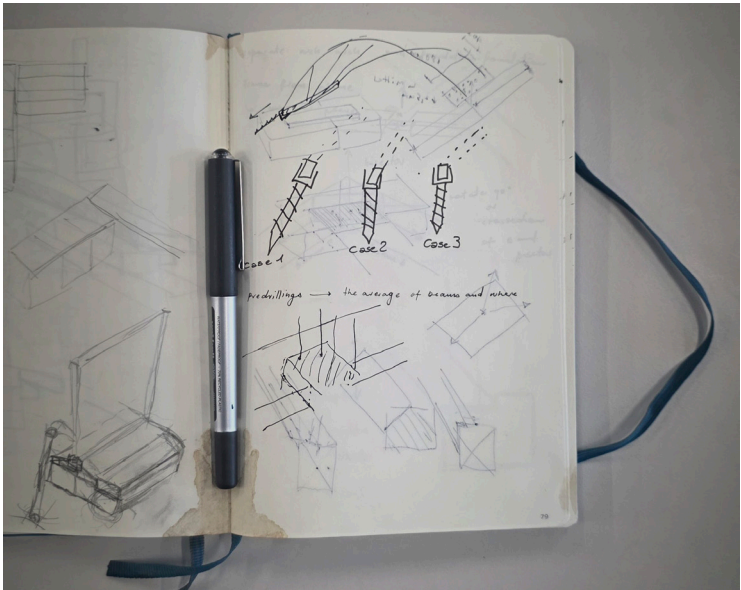
6 // TIMBER MODEL CREATION

Timber beams and joints are generated from the final RF system. Cross-section dimensions and joint milling depths are parametrized independently for the arch, foundation, and inner beam categories.

Grasshopper Full Project







Detailing: Group B work

Revision

Detailing

Fabrication

Assembly

Fabrication: Group C work

Revision

Detailing

Fabrication

Assembly

Assembly: Group D work

6 / Built Pavilion

Images of finished project

text

Conclusion

Conclusion clean text

Credits

Gramazio Kohler Research
Institute of Technology in Architecture (ITA)
Department of Architecture (D-ARCH)
ETH Zürich

HARDLEAF

A design-build research project developed through Coding Architecture II (FS26)

Project Partners

Verein Quartiergarten Hard Zürich
Bergmeister + Partner AG

Community Partners

David Appel
Jesús Medina
Verein Quartiergarten Hard Zürich

Teaching Team

Fabio Gramazio
Lorin Wiedemeier
Gonzalo Casas
Eric Gozzi

Students

Paola Ak
Tabea Brochier
Marco Brönimann
Angelina Derzhitskaya-Zeman
Anneke Iten de León
Lisa Kammerick

Aron Karrer
Žan Kočunik
Chia-Wei Lee
Alessia Martino
Cami Minger
Rachele Pasteris

Lucija Ratkić
Irene Schnellmann
Gilda Varini
Yiwen Wang
Florian Weinke
Wanru Zhao

Structural Engineering

Bergmeister + Partner AG

Fabrication Support

Panos Papacharalambous

Matthias Helmreich

Nico [.....]

Coordination

Lorin Wiedemeier

Gonzalo Casas

Design Concept

Timo Feddern

Alessia Martino

Žan Kočunik

Construction and Assembly

Teaching Team and Students of Coding Architecture II FS26

Special Thanks

Panos Papacharalambous

Matthias Helmreich

David Appel

Jesús Medina

and all members of Quartiergarten Hard Zürich who supported the project throughout its development and construction.

